Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

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1. Problem Statement (Problem to be solved)

Libraries face difficulties in managing books, student records, and issue/return processes manually. This leads to errors, data duplication, time consumption, and lack of real-time tracking of book availability.

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2. Idea / Solution Description

The proposed solution is to develop a centralized web-based Smart Library system. It will manage book records, student details, issue and return tracking, and due dates. The system will provide real-time updates and easy access for both librarians and students.

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3. Novelty / Uniqueness

The system includes features such as real-time book availability, role-based access control (librarian and student), automated issue/return tracking, and simple dashboard views. These features reduce manual work and improve efficiency.

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4. Social Impact / Customer Satisfaction

This solution improves library efficiency and reduces manual workload. Students can easily check book availability, and librarians can manage records quickly. It ensures accuracy and saves time, leading to better user satisfaction.

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5. Business Model (Revenue Model)

The system can follow a subscription-based model where schools, colleges, and libraries pay monthly or yearly fees. Additional revenue can come from premium features like analytics, reporting, and customization.

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6. Scalability of the Solution

The solution is scalable and can be used by small libraries as well as large institutions. With cloud-based deployment and modular design, it can handle increasing users and data efficiently.